3.9 Mathematical Requirements

In order to develop their skills, knowledge and understanding in science, candidates need to have been taught, and to have acquired competence in, the appropriate areas of mathematics relevant to the subject as set out below.

	Candidates should be able to:
Arithmetic and computation	• recognise and use expressions in decimal and standard form • use ratios, fractions and percentages • use calculators to find and use x^n, $1/x$, $\sqrt{x}$, $\log_{10}x$, e^x, $\log_e x$ • use calculators to handle sin x, cos x, tan x when x is expressed in degrees or radians.
Handling data	• use an appropriate number of significant figures • find arithmetic means • make order of magnitude calculations.
Algebra	• understand and use the symbols: =, <, <<, >>, >, $\propto$, ~. • change the subject of an equation by manipulation of the terms, including positive, negative, integer and fractional indices • substitute numerical values into algebraic equations using appropriate units for physical quantities • solve simple algebraic equations.
Graphs	• translate information between graphical, numerical and algebraic forms • plot two variables from experimental or other data • understand that $y = mx + c$ represents a linear relationship • determine the slope and intercept of a linear graph • draw and use the slope of a tangent to a curve as a measure of rate of change • understand the possible physical significance of the area between a curve and the x -axis and be able to calculate it or measure it by counting squares as appropriate • use logarithmic plots to test exponential and power law variations • sketch simple functions including $y = k/x$, $y = kx^2$, $y = k/x^2$, $y = \sin x$, $y = \cos x$, $y = e^{-kx}$
Geometry and trigonometry	• calculate areas of triangles, circumferences and areas of circles, surface areas and volumes of rectangular blocks, cylinders and spheres • use Pythagoras' theorem, and the angle sum of a triangle • use sines, cosines and tangents in physical problems • understand the relationship between degrees and radians and translate from one to the other.